



Taskset

The taskset command is used to set or retrieve the CPU affinity of a running process given its pid, or to launch a new command with a given CPU affinity.

Getting Started

Usage

The default behavior is to run a new command with a given affinity mask

```
$ taskset [mask] [command] [arguments]
```

Usage #2

```
$ taskset -p [pid]
```

```
$ taskset -p [mask] [pid]
```

You can also retrieve the CPU affinity of an existing task

Common Options

Command	Alternavite	Meaning
-a	--all-tasks	Set or retrieve the CPU affinity of all the task (threads) for a given PID.
-c	--cpu-list	Interpret mask as numerical list of processors instead of a bitmask. Numbers are separated by commas and may include ranges. For example: 0,5,8-11
-p	--pid	Operate on an existing PID and do not launch a new task.
-h	--help	Display help text and exit.

Command	Alternavite	Meaning
<code>-v</code>	<code>--version</code>	Print version and exit.

Example

View CPU Affinity

```
$ taskset -cp 29523
pid 29523's current affinity list: 0-15
```

That mean process with PID **29523** is active on CPU range from **[0,1,...,15]** which means **16** total.

PID: Process Identifier - is a unique numerical identifier given to each running process, you can view PID of processes in **top** or **htop** tools

If you run the above command but get the message **bad usage**, try with PID = 1

```
$ taskset -cp 1
```

Set CPU Affinity

Now let's try to set process in to a specify CPU. For example, we have a process has PID = 14846

Let's see the CPU affinity list of this process first:

```
$ taskset -cp 14846
pid 14846's current affinity list: 0-15
```

Change CPU affinity to 1:

```
$ taskset -cp 1 14846
pid 14846's current affinity list: 0-15
pid 14846's new affinity list: 1
```

As we see, CPU affinity was changed

Set CPU Affinity with muliple value

That mean new CPU affinity list is **[0,1,2,3]**

```
$ taskset -cp 0-3 14846
pid 14846's current affinity list: 1
pid 14846's new affinity list: 0-3
```

You can set single CPU with the following command:

```
$ taskset -cp 5,8,12 14846
pid 14846's current affinity list: 0-3
pid 14846's new affinity list: 5,8,12
```

That mean new CPU affinity list is **[5,8,12]**

Other

You can set affinity by range with other like a specify CPU

```
$ taskset -cp 1-3,12 14846
```

Or

```
$ taskset -cp 1-6:2 14846
```

The suffix ":N" specifies stride in the range, for example 0-10:3 is interpreted as 0,3,6,9 list.

Also see

[taskset — Linux manual page \(man7.org\)](#)

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